Finite Difference Solution of the 2D Poisson Equation: grid refinement

The problem

Consider the Poisson equation

$$-\nabla^2 u(x,y) = 2\pi^2 \sin(\pi x)\sin(\pi y) = f(x,y),$$

defined in the domain

$$\Omega = [0,3] \times [0,3],$$

with homogeneous Dirichlet boundary conditions

$$u(x,y) = 0 \quad \text{for} \quad (x,y) \in \partial\Omega.$$

The exact analytical solution is

$$u(x,y) = \sin(\pi x)\sin(\pi y).$$

Discretization

Introduce a uniform grid $i, j = 0, 1, \dots N$:

$$x_i = i\Delta x, \quad y_j = j\Delta x, \quad \Delta x = \frac{3}{N}.$$

The number of grid points is a power of 2: $N = 2^n$. In the implementation: we take as large n as possible for a reasonable calculation time. The calculation should not last more than $\simeq 2$ minutes.

The Laplacian is approximated using the 5–point stencil:

$$\frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta x^2} = -f_{i,j}.$$

The quality of solution and convergence of the iteration will be monitored with the ”energy” functional:

$$\mathcal{S}[u] = \int_\Omega \left[\frac{1}{2}(\nabla u)^2 - fu\right] d\Omega.$$

The analytical value for the exact solution is $\mathcal{S} = -(3\pi/2)^2$.

Relaxation

Implement the point relaxation (Gauss–Seidel). A single iteration consists in scanning the grid and replacing the potential in point (i,j) (besides the boundary) by a new value

$$u_{i,j} := \frac{1}{4}\left(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - \Delta x^2 f_{i,j}\right).$$

To report:

1. the convergence of the iteration
2. how the residuum $\delta_{ij} = \frac{u_{i+1,j}+u_{i-1,j}+u_{i,j+1}+u_{i,j-1}-4u_{i,j}}{\Delta x^2} + f_{i,j}$ changes during the iteration
3. convergence rate in terms of $\mathcal{S}$ vs N
4. u once convergence is reached

Overrelaxation

We change the iteration procedure to

$$u_{i,j} := \frac{w}{4}\left(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - \Delta x^2 f_{i,j}\right) + (1-w)u(i,j).$$

To report:

1. search for optimal w
2. the convergence for optimal w and its rate with respect to N

Grid refinement

We start the iteration from a grid with $N_0 = 2^k$ with $k = n-5$. Once the convergence is reached we prepare the calculation on a finer grid with $N_1 = 2^{k+1}$. We keep the u values in the grid nodes that were covered with the grid of $N_0 \times N_0$ points. In the new nodes we set at the start the average of the nearest nodes of the old grid (2 or 4 depending on the position of the node on the finer grid). We proceed till the solution on the destination grid is obtained.

To report:

1. how large can be n now?
2. $\mathcal{S}$ as a function of the total number of iterations for all the grids covered